Higher coursework assessment task

Qualifications Scotland Teisteanasan Alba

Higher Engineering Science
Assignment
Finalised Marking Instructions

These marking instructions have been prepared by examination teams for use by Qualifications Scotland appointed markers when marking external course assessments.

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General marking principles

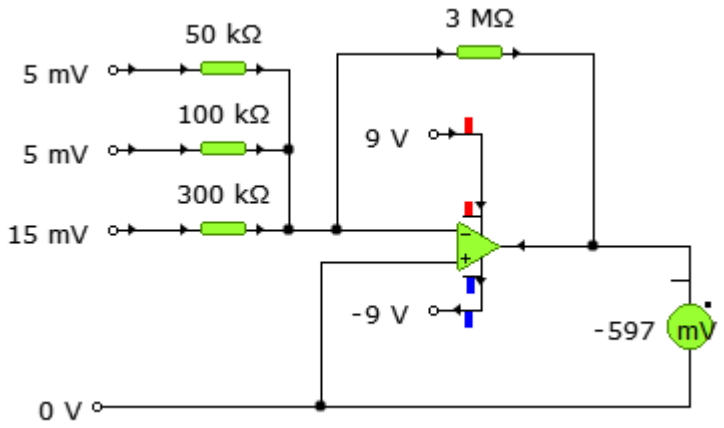
This information is provided to help you understand the general principles that must be applied when marking candidate responses in this assignment. These principles must be read in conjunction with the detailed/specific marking instructions, which identify the key features required in candidate responses.

a Marks for each candidate response must always be assigned in line with these general marking principles and the specific marking instructions for this assessment.

b Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

c If a specific candidate response is not covered by either the general marking principles or detailed marking instructions, you must seek guidance from your team .

Detailed marking instructions

Task			Expected answer(s)	Max mark	Additional guidance
1	a			2	**1 mark** for correct positioning and connections of all components. **1 mark** for correct values of all resistors and voltage supplies. Accept variable voltage inputs at the correct voltages as suitable replacements for the illustrated fixed voltage inputs.

<table>
<tr><th colspan="3">Task</th><th>Expected answer(s)</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>1</td><td>b</td><td></td><td>
<table>
<tr><th>Actual result</th><th>Amendments made</th></tr>
<tr><td>-300 mV</td><td>Additional inverting op-amp with gain of -1 added.</td></tr>
<tr><td>+150 mV</td><td>Change V_2 input resistor to 50 kΩ.</td></tr>
<tr><td>+750 mV</td><td>Change V_3 input resistor to 150 kΩ.</td></tr>
</table>
</td><td>6</td><td>Simulated circuits may produce output voltages slightly different from expected results. Take this into account when marking.
Apply FTE from circuit produced in Task 1a.

1st actual result:
1 mark for-300 mV.

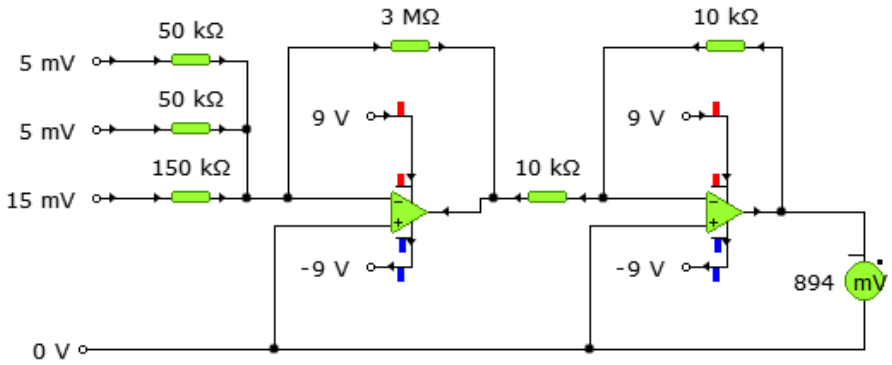
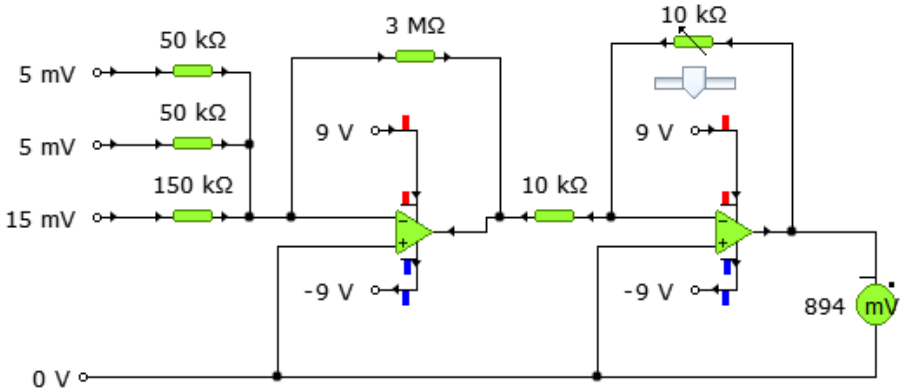
1st amendment:
1 mark for stating addition of an inverting op-amp with gain of -1.
Apply FTE from 1st actual result.

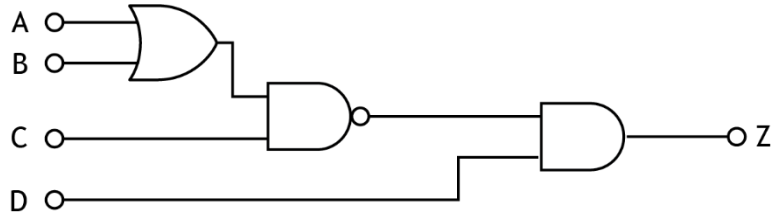
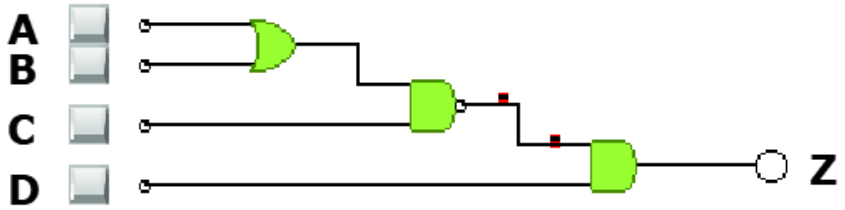
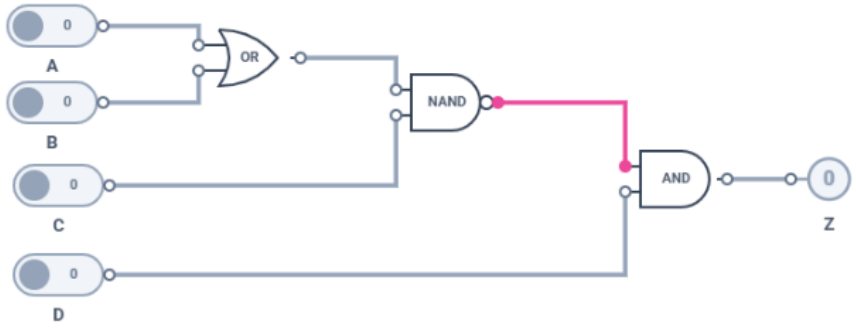
2nd actual result:
1 mark for+150 mV.
Apply FTE from 1st amendment.

2nd amendment:
1 mark for V_2 input resistor changing to 50 kΩ.
Apply FTE from 2nd actual result.

3rd actual result:
1 mark for +750 mV.
Apply FTE from 2nd amendment.

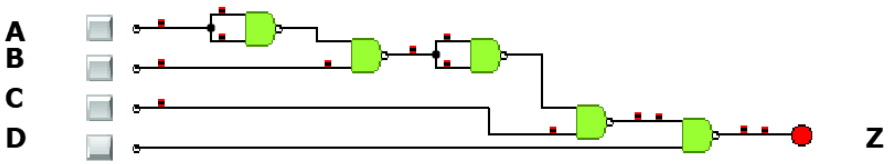
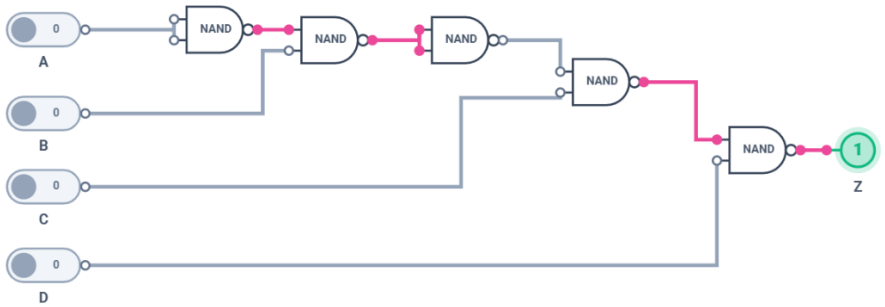
3rd amendment:
1 mark for V_3 input resistor changing to 150 kΩ.
Apply FTE from 3rd actual result.</td></tr>
</table>

Task			Expected answer(s)	Max mark	Additional guidance
1	c			2	**1 mark** for correctly connected inverting op-amp with a gain of -1. **1 mark** for correctly changing input resistor values for V_2 and V_3. Apply FTE from Task 1b.
	d			2	**1 mark** for variable resistor. **1 mark** for correct positioning of the variable resistor. Accept alternatives that produce the desired effect, such as replacing the 3 MΩ resistor or the 10 kΩ input resistor with a suitable variable resistor.

Task			Expected answer(s)	Max mark	Additional guidance
2	a			3	**1 mark** for A **or** B connected correctly. **1 mark** for NAND (or AND and NOT) connected correctly. **1 mark** for AND connected correctly. No marks awarded if simulated or constructed.
	b		**Example 1**  **Example 2** 	2	**1 mark** for suitable input and output devices. **1 mark** for correct simulation or construction of logic circuit from Task 2a.

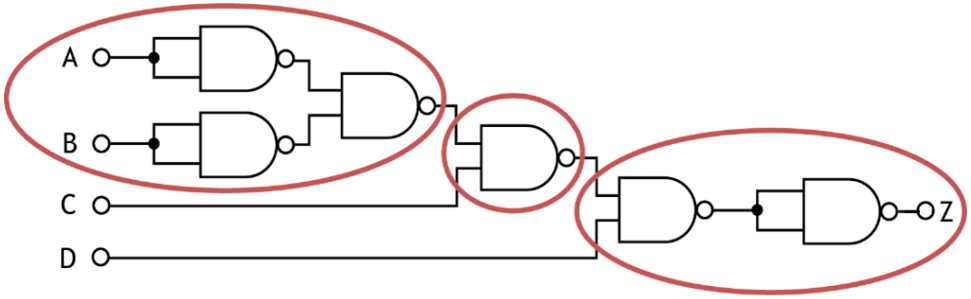
<table>
<tr><th colspan="3">Task</th><th>Expected answer(s)</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>2</td><td>c</td><td></td><td>
<table>
<tr><th>A</th><th>B</th><th>C</th><th>D</th><th>Z</th><th>NAND equivalent</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td></td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td></td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td></td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td></td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td></td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td></td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td></td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td></tr>
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<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td></td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td></td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td></td></tr>
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<tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td></td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td></td></tr>
</table>
</td><td>1</td><td>1 mark for column Z correct.

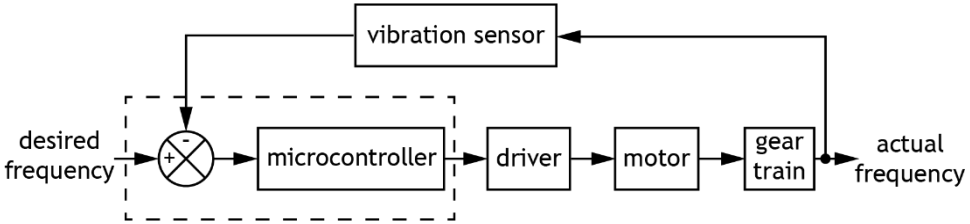
Apply FTE from circuit in Task 2b.</td></tr>
</table>

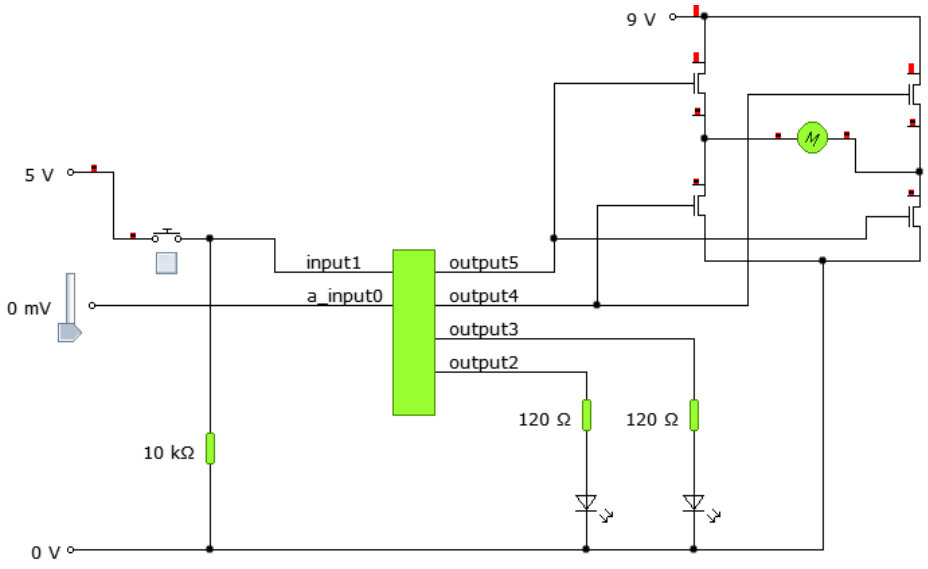
Task			Expected answer(s)	Max mark	Additional guidance
2	d		**Example 1**  **Example 2**  	1	**1 mark** for correct simulation or construction. Input and output devices must be included.

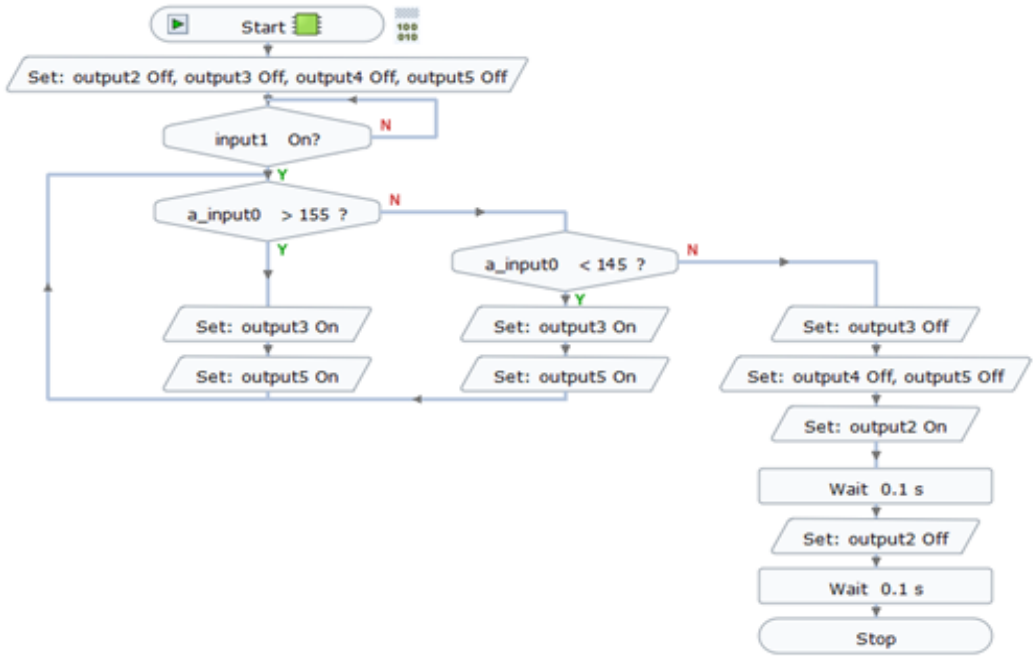
Task			Expected answer(s)	Max mark	Additional guidance
2	e			1	**1 mark** for NAND equivalent column correct. Apply FTE from Task 2d.

A	B	C	D	Z	NAND equivalent
0	0	0	0	0	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	1	1	0
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	0	1
0	1	1	1	0	1
1	0	0	0	0	1
1	0	0	1	1	0
1	0	1	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	1	0	0

Task			Expected answer(s)	Max mark	Additional guidance
2	f			3	**1 mark** for correct OR equivalent connected correctly. **1 mark** for NAND connected correctly. **1 mark** for correct AND equivalent connected correctly. Do not accept simulation or construction evidence.
	g		The original circuit would require three ICs to be constructed whereas the NAND equivalent circuit would only need two. As there are fewer ICs, the cost would be reduced. Fewer ICs reduces manufacturing time due to less complex circuitry.	2	**1 mark** for identifying that fewer ICs would be required. **1 mark** for a suitable comment relating to cost or manufacture. Accept any other correctly described advantages or disadvantages.

Task			Expected answer(s)	Max mark	Additional guidance
3	a			5	**1 mark** for error detector with negative feedback configuration and dashed control boundary. **1 mark** for microcontroller/ control unit in correct order. (do not accept 'control' on its own or 'op-amp'). **1 mark** for driver, motor and gear (train) in correct order. **1 mark** for vibration (frequency) sensor in correct feedback position. **1 mark** for feedback loop with correct direction arrowheads (both lines).

Task			Expected answer(s)	Max mark	Additional guidance
3	b			6	**1 mark** for digital input connected correctly to input 1 with 5 V supply. **1 mark** for variable voltage analogue input connected correctly to input 0. **1 mark** for LED outputs with 120 Ω resistors connected correctly to outputs 2 and 3. **1 mark** for MOSFETs connected correctly to outputs 4 and 5, power supplies, and motor.

Task			Expected answer(s)	Max mark	Additional guidance
3	b	(continued)			
				6	**2 marks** for fully correct flowchart. **1 mark** for maximum two errors. Allow FTE from circuit.

<table>
<tr><th colspan="3">Task</th><th>Expected answer(s)</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr>
<td>3</td>
<td>c</td>
<td></td>
<td>
<table>
<tr><th>Actual result</th><th>Amendments</th></tr>
<tr><td>Red LED and motor switch on.</td><td>No amendments.</td></tr>
<tr><td>Motor switches on and turns in the same direction.

Red LED switches on.</td><td>Modify flowchart so that output 4 switches on and motor turns in the opposite direction.</td></tr>
<tr><td>When input voltage is increased, the green LED flashes once, and the motor slows to a halt.

Flowcharts stops.</td><td>Additon of a loop to make the green LED flash.

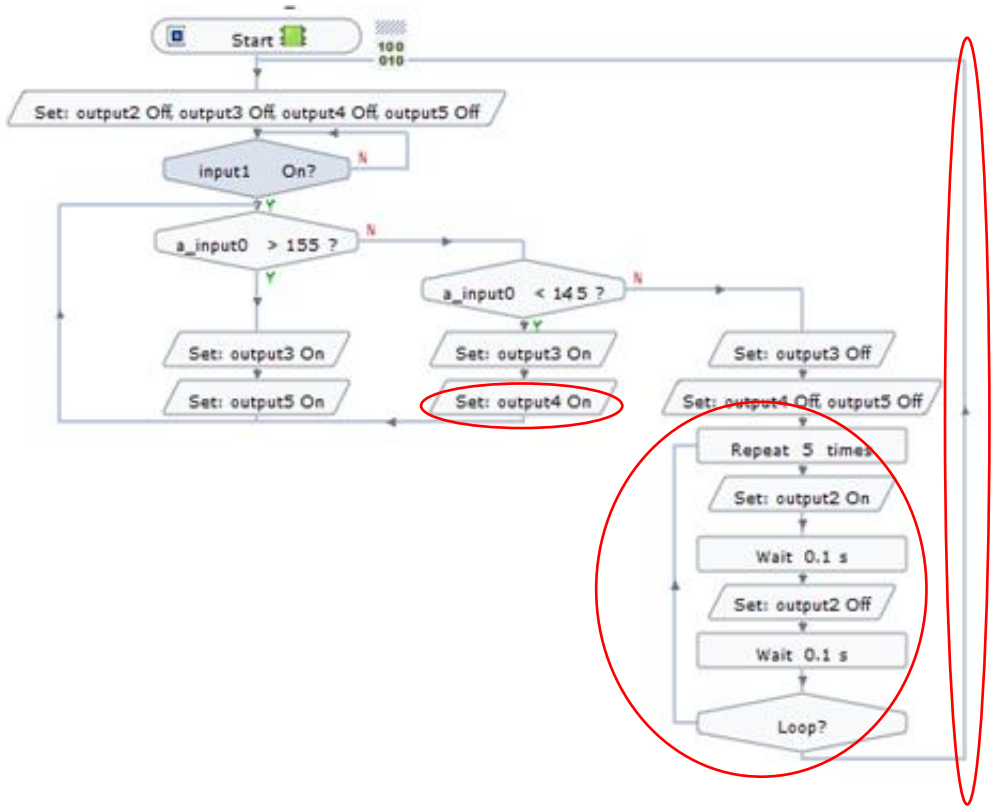
‘stop’ replaced with a loop to the start.</td></tr>
</table>
</td>
<td>5</td>
<td>1st actual result:
1 mark for both red LED and motor switching on.

2nd actual result:
1 mark for motor switches on and turning in same direction and red LED switches on.

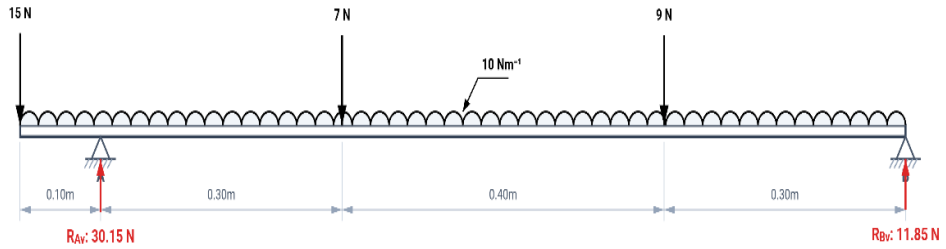
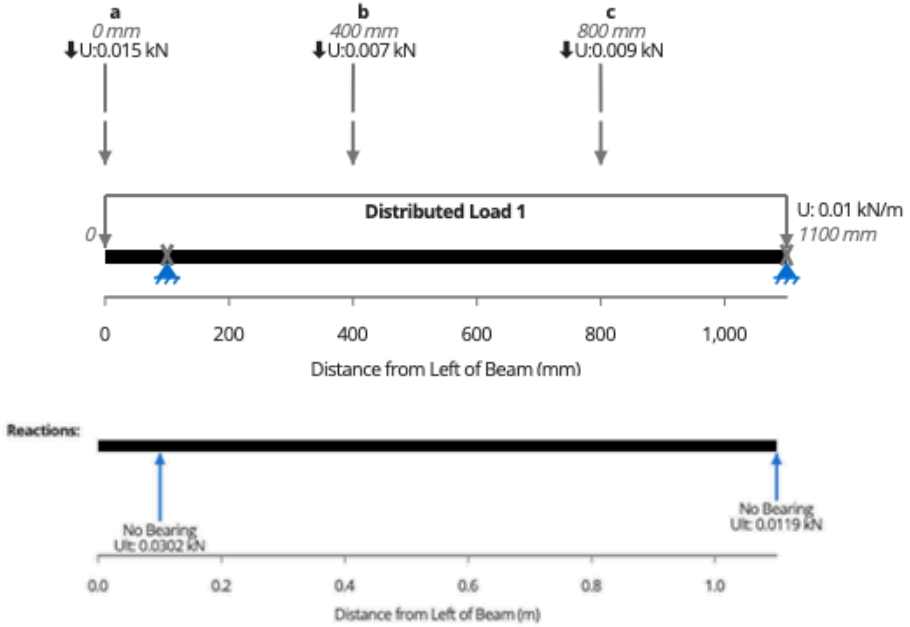
2nd amendment:
1 mark for modifying flowchart to switch output 4 on.

3rd actual result:
1 mark for the green LED flashing once, motor stops, and no loop to start.

3rd amendment:
1 mark for addition on 5x loop to make green LED flash and ‘stop’ replaced with a loop to start.</td>
</tr>
</table>

Task			Expected answer(s)	Max mark	Additional guidance
3	d			1	**1 mark** for all correct amendments. Apply FTE from Task 3c.

Task			Expected answer(s)	Max mark	Additional guidance
3	**e**			**5**	All comments must be evaluative, reflect the entry in the yes/ no column, and be based on test results and amendments from Task 3c/ Task 3d.
			i - The system does not start until the start switch is pressed.		**1 mark** for points i and ii.
			ii - When the reading was above or below the desired range, the red LED switched on.		
			iii - When the reading was above the desired range, the motor turned.		**1 mark** for point iii.
			iv - When the reading was lower than the desired range, the motor turned in the opposite direction (due to amendment made).		**1 mark** for point iv.
			v - When the reading was within the desired range, the motor stopped, the red LED switched off, and the green LED flashed five times over one second (due to amendment made).		**1 mark** for point v.
			vi - After the green LED flashed, the system returned to the start ready for the next press of the start switch.		**1 mark** for point vi.

Task			Expected answer(s)	Max mark	Additional guidance
4	a		**Example 1**  **Example 2**  	2	**1 mark** for correct simulation, including forces and distances. **1 mark** for correct reaction forces. A= 30.15 N, B= 11.85 N. Apply FTE from candidate simulation. Note: reaction forces rounding will be determined by software used.

Task			Expected answer(s)	Max mark	Additional guidance
4	b		Support A is not strong enough for the situation. Support B is strong enough.	1	**1 mark** for correctly stating the suitability of both supports. Apply FTE from Task 4a.

[END OF MARKING INSTRUCTIONS]